

Part 3: Multiple-Variable Calculus

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Recall $\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$.

Vector space structure:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2),$$

$$k(x_1, y_1) = (kx_1, ky_1), \quad k \in \mathbb{R}, (x_1, y_1) \in \mathbb{R}^2.$$

Distance between two points (x_1, y_1) and (x_2, y_2) :

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

Example 1

- **Closed disk** $\overline{B}(x, \varepsilon) = \{M \in \mathbb{R}^n \mid d(M, x) \leq \varepsilon\}$.
- **Open disk** $B(y, \delta) = \{M \in \mathbb{R}^n \mid d(M, y) < \delta\}$.

Definition 1

A **function of two variables** $f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is a rule that assigns $M(x, y) \in D$ to a unique value $z = f(M) = f(x, y)$.

D is called **the domain** of f . The **range** is the set $\{f(M) \mid M \in D\}$.

x, y : independent variables; z : dependent variable.

We have various ways of looking at f :

- as a function of 2 variables (x, y) ;
- as a function of a single point $M(x, y)$.

By the same way, we can define a function of three or more variables.

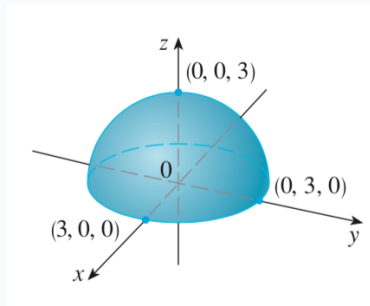
Example 2

Find the domains of the following functions:

1 $z = z(x, y) = \frac{x}{\sqrt{1-x^2-y^2}}.$

2 $u = f(x, y, z) = \arcsin \sqrt{x^2 + y^2}.$

- The graph of $z = f(x, y): D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ is the set $\Gamma(f) = \{(x, y, f(x, y)), (x, y) \in D\} \subset \mathbb{R}^3$.
- The graph of $z = f(x, y)$ is a **surface** in $\mathbb{R}^3$.



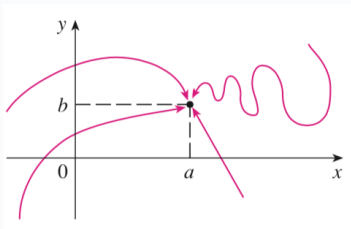
Hình: $z = \sqrt{9 - x^2 - y^2}$

Definition 1.1

Let $f: D \rightarrow \mathbb{R}$, D contains an open disk $B(M_0, \varepsilon)$, possibly except $M_0(x_0, y_0)$. We say that the *limit of $f(x, y)$ is L* when $M(x, y)$ tends to $M_0(x_0, y_0)$ iff

$$\lim_{M \rightarrow M_0} f(M) = L \Leftrightarrow \forall \varepsilon > 0, \exists \delta(\varepsilon) > 0 : 0 < d(M, M_0) < \delta \\ \Rightarrow |f(M) - L| < \varepsilon.$$

In the line, $x \rightarrow x_0$ either from the left or the right.



Note: If the limit exists, then $f(M)$ must approach same L no matter how M approaches M_0 .

To show **nonexistence** of the limit:

$f(M) \rightarrow a_1$ as $M \rightarrow M_0$ along the path C_1 , $f(M) \rightarrow a_2$ as $M \rightarrow M_0$ along the path C_2 ; but $a_1 \neq a_2$.

Example 3

Compute the limits

$$1 \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - 2y^2}{x^2 + y^2}$$

$$2 \quad \lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4}$$

$$3 \quad \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + 2xy - 3y^2}{x^2 + 2y^2}$$

1 Let $(x, y) \rightarrow (0, 0)$ along the line $y = kx$:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - 2y^2}{x^2 + y^2} = \lim_{x \rightarrow 0} \frac{x^2 - 2k^2x^2}{x^2 + k^2x^2} = \frac{1 - 2k^2}{1 + k^2} \text{ depends on } k.$$

The limit is not existent.

2 Let $(x, y) \rightarrow (0, 0)$ along the parabola $x = ky^2$:

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2 + y^4} = \lim_{y \rightarrow 0} \frac{ky^4}{k^2y^4 + y^4} = \frac{k}{1 + k^2} \text{ depends on } k.$$

The limit is not existent.

Limit of a function of several variables has similar properties as those of limit of a function of a single variable: uniqueness, order preserving,...

Theorem 1.1

Assume that $\lim_{M \rightarrow M_0} f(M)$, $\lim_{M \rightarrow M_0} g(M)$ are finite.

$$\lim_{M \rightarrow M_0} (f(M) \pm g(M)) = \lim_{M \rightarrow M_0} f(M) \pm \lim_{M \rightarrow M_0} g(M)$$

$$\lim_{M \rightarrow M_0} f(M) \cdot g(M) = \lim_{M \rightarrow M_0} f(M) \cdot \lim_{M \rightarrow M_0} g(M).$$

Example 4

Compute the limit $\lim_{(x,y) \rightarrow (2,1)} \frac{x^3 - y^3}{x^2 + y^2}.$

Theorem 1.2 (Squeeze theorem)

If $f(M) \leq g(M) \leq h(M)$ when $M \in B(M_0, \varepsilon) \setminus \{M_0\}$ and

$$\lim_{M \rightarrow M_0} f(M) = \lim_{M \rightarrow M_0} h(M) = L,$$

then

$$\lim_{M \rightarrow M_0} g(M) = L.$$

Example 5

Compute the limits

$$1 \quad \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} \cos \frac{1}{xy}$$

$$2 \quad \lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^2 + y^2}$$

$$1 \quad \text{We have } 0 \leq \left| \sqrt{x^2 + y^2} \cos \frac{1}{xy} \right| \leq \sqrt{x^2 + y^2} \xrightarrow{x,y \rightarrow 0} 0.$$

$$\text{By Squeeze theorem, } \lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} \cos \frac{1}{xy} = 0.$$

$$2 \quad \text{We have } 0 \leq \left| \frac{2x^2y}{x^2 + y^2} \right| = \left| 2y \frac{x^2}{x^2 + y^2} \right| \leq 2|y| \xrightarrow{x,y \rightarrow 0} 0.$$

$$\text{By Squeeze theorem, } \lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^2 + y^2} = 0.$$

Definition 2

Let $f(M)$ be defined on D , $M_0 \in D$, D contains an open disk centered at M_0 . The function $f(M)$ is said to be **continuous at M_0** iff $\lim_{M \rightarrow M_0} f(M) = f(M_0)$.

$f(M)$ is said to be **continuous on D** if it is continuous at any $M_0 \in D$.

Example 6

Investigate the continuity at $(x, y) = (0, 0)$ of the following functions

$$\textcircled{1} \quad f(x, y) = \begin{cases} \cos \frac{2xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

$$\textcircled{2} \quad g(x, y) = \begin{cases} \frac{x^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Definition 3

Let $f(x, y) : D$

The *partial derivative with respect to x* is

$$\begin{aligned} f'_x(x_0, y_0) &= \frac{\partial f}{\partial x}(x_0, y_0) = \frac{d}{dx} f(x, y_0) \Big|_{x=x_0} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} \end{aligned}$$

The *partial derivative with respect to y* is

$$\begin{aligned} f'_y(x_0, y_0) &= \frac{\partial f}{\partial y}(x_0, y_0) = \frac{d}{dy} f(x_0, y) \Big|_{y=y_0} \\ &= \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} \end{aligned}$$

Rule: when differentiating with respect to x , y is regarded as a constant; when differentiating with respect to y , x is regarded as a constant.

Example 7

Compute the partial derivatives of the following functions

❶ $f(x, y) = x^y$ at $(2, 1)$.

❷ $u(x, y, z) = z\sqrt{x^2 + y^2}$.

❸
$$f(x, y) = \begin{cases} \frac{x^3 - 3xy^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

The chain rule (case 1)

Theorem 1.3

Suppose that $z = f(x, y)$ has continuous partial derivatives, where $x = x(t)$ and $y(t)$ are both differentiable functions of t . Then $z = f(x(t), y(t))$ is a differentiable function of t and

$$z'(t) = f'_x \cdot x'(t) + f'_y \cdot y'(t).$$

Example 8

$$\begin{aligned} z &= \sqrt{1 + x^4 + y^2}, \quad x = \ln t, \quad y = \sin t, \quad \text{then} \\ \frac{dz}{dt} &= -\sin(x + 2y) \frac{1}{t} - 2 \sin(x + 2y) \cos t \\ &= -\sin(\ln t + 2 \sin t) \left(\frac{1}{t} + 2 \cos t \right). \end{aligned}$$

Recall: The composition $g \circ f : X \subset \mathbb{R}^2 \xrightarrow{f} Y \subset \mathbb{R}^2 \xrightarrow{g} \mathbb{R}$ is

$$(x, y) \xrightarrow{f} (u(x, y), v(x, y)) \xrightarrow{g} \underbrace{g((u(x, y), v(x, y)))}_{F(x, y)}$$

Theorem 1.4

Assume that $g(u, v)$ has continuous partial derivatives in Y , u, v have partial derivatives in X . Then the composition function $F = g \circ f : X \rightarrow \mathbb{R}$ has partial derivatives in X and

$$F'_x = g'_u \cdot u'_x + g'_v \cdot v'_x, \quad F'_y = g'_u \cdot u'_y + g'_v \cdot v'_y$$

Definition 4

i) Given $f(x, y): D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$, $M_0(x_0, y_0) \in D$. If we can express

$$\Delta f = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = A\Delta x + B\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y,$$

where the constants A, B depend only on x_0, y_0 , and $\varepsilon_1 \rightarrow 0, \varepsilon_2 \rightarrow 0$ as $\Delta x, \Delta y \rightarrow 0$, we say that $f(x, y)$ is differentiable at M_0 . Notice that $\varepsilon_1, \varepsilon_2$ are functions of Δx and Δy .

ii) $df(x_0, y_0) = A\Delta x + B\Delta y$ is called the total differential of f at M_0 .

iii) f is said to be differentiable on D if f is differentiable at all $M_0 \in D$.

Example 9

Is the function $z = 2x - y^2$ differentiable at $(1, 0)$?

- If f is differentiable at M_0 then f is continuous at M_0 .

Indeed, let $\Delta x, \Delta y \rightarrow 0$, then $\Delta f = A\Delta x + B\Delta y + \varepsilon_1\Delta x + \varepsilon_2\Delta y \rightarrow 0$.

- $\Delta y = 0$, $\frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} = A + \varepsilon_1$.

Passing to the limit as $\Delta x \rightarrow 0$: $A = f'_x(x_0, y_0)$.

Similarly, $B = f'_y(x_0, y_0)$. $df = f'_x\Delta x + f'_y\Delta y$.

If f is differentiable at M_0 then f has partial derivatives at M_0 . However, **the converse is not necessary true**.

Take $f(x, y) = x$, then $\Delta x = dx$.

Similarly, $\Delta y = dy$, so

$$df = f'_x dx + f'_y dy$$

Example 10

$$\text{Is } f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0), \end{cases} \text{ differentiable at } (0, 0)?$$

f is not differentiable at $(0, 0)$ although $f'_x(0, 0) = f'_y(0, 0) = 0$.

Theorem 1.5 (Sufficient condition for differentiability)

If $f'_x(x, y), f'_y(x, y)$ exist in $B(M_0, \varepsilon)$ and are continuous at M_0 . Then, $f(x, y)$ is differentiable at M_0 and

$$df(M_0) = f'_x(M_0)dx + f'_y(M_0)dy.$$

Example 11

Compute the total differential of the following functions

$$a) z = \frac{1}{2}(x^2 + y^2)$$

$$b) z = x^y$$

$$c) z = \arctan xy$$

$$d) u = \frac{z}{\sqrt{x^2 + y^2}}, du(1, 2, 3).$$

Recall: linear approximation

$$\Delta f = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \approx \underbrace{f'_x(x_0, y_0)\Delta x + f'_y(x_0, y_0)\Delta y}_{df(x_0, y_0)}$$

Implicit function $z = z(x, y)$ determined by $f(x, y, z) = 0$

Given $x^2 + y^2 + z^2 = 9$, we can rewrite $z = \pm\sqrt{9 - x^2 - y^2}$.

Theorem 1.6

Let $f(x, y, z): D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ have continuous partial derivatives in a neighborhood of $M_0(x_0, y_0, z_0)$, $f(M_0) = 0$ and $f'_z(M_0) \neq 0$. Then, the equation $f(x, y, z) = 0$ determines in a neighborhood of (x_0, y_0) an implicit function $z = z(x, y)$ such that $z(x_0, y_0) = z_0$ and z has continuous partial derivatives. Moreover,

$$z'_x(x_0, y_0) = -\frac{f'_x(M)}{f'_z(M)}, \quad z'_y(x_0, y_0) = -\frac{f'_y(M)}{f'_z(M)}.$$

Example 12

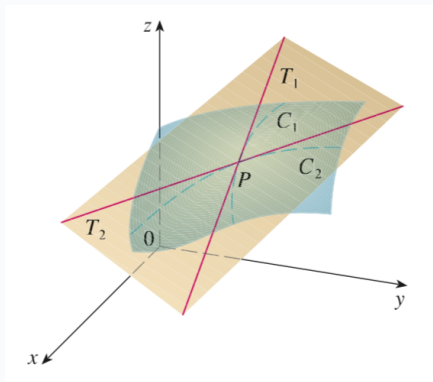
- ❶ Let $z = z(x, y)$ be determined by the equation $2x^3y - x^2 + xe^z + 2xyz = 2$. Compute $z(1; 1)$, $z'_x(1; 1)$, $z'_y(1; 1)$.
- ❷ Let $z = z(x, y)$ be determined by the equation $z^3 + x^2y + xy^2 = 3xyz$.

❶ $f(1; 1; z(1, 1)) = 0 \Leftrightarrow e^z + 2z = 1, z(1, 1) = 0.$

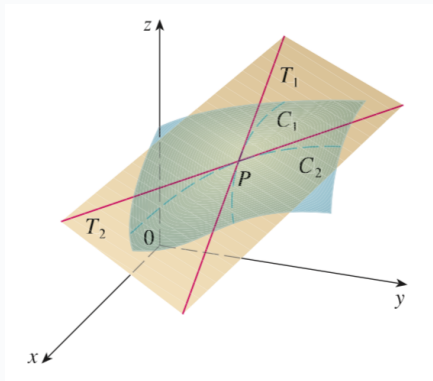
$$z'_x = -\frac{6x^2y - 2x + e^z + 2yz}{xe^z + 2xy}, z'_x(1; 1) = -\frac{5}{3};$$

$$z'_y = -\frac{2x^3 + 2xz}{xe^z + 2xy}, z'_y(1; 1) = -\frac{2}{3}.$$

Consider a surface $S : z = f(x, y)$, where f has continuous first partial derivatives. $P(x_0, y_0, z_0) \in S$.



Let C_1, C_2 be two curves obtained by intersecting the planes $y = y_0$ and $x = x_0$ with S at P . Let T_1 and T_2 be the tangent lines to C_1 and C_2 at P . The plane contains T_1 and T_2 is called **the tangent plane** to the surface S at P .



$$C_1 : \begin{cases} y = y_0 \\ z = f(x, y_0) \end{cases} \Rightarrow T_1 : \begin{cases} y = y_0 \\ z - z_0 = f'_x(x_0, y_0)(x - x_0). \end{cases}$$

A directional vector of the tangent plane $\vec{u}_1 = (1; 0; f'_x(P))$. Similarly, $\vec{u}_2 = (0; 1; f'_y(P))$. Hence, a normal vector is $\vec{n} = \vec{u}_1 \times \vec{u}_2 = (-f'_x(P); -f'_y(P); 1)$.

An equation of the tangent plane to the surface $S : z = f(x, y)$ at the point $P(x_0, y_0, z_0)$ is

$$z - z_0 = f'_x(P)(x - x_0) + f'_y(P)(y - y_0)$$

Example 13

Write an equation of the tangent plane at $M(1; 1; 3)$ of the surface $z = x^2y + 2xy^2$.

Suppose that the first order partial derivative of f are continuous. Then for (x, y) close to (x_0, y_0) , we have the linear approximation

$$f(x, y) \approx f(x_0, y_0) + f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0).$$

Example 14

$f(x, y) = x^3 - 2xy^2$. Approximate $f(2.01; 1.98)$.

Example 15

Approximate $A = \sqrt[3]{1,02^2 + 1,98^3 - 1}$; $B = \ln(\sqrt[3]{1,02} + \sqrt[4]{0,98} - 1)$.

Definition 5

Second order partial derivatives of $f(x, y)$ are:

$$f''_{xx} = \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$$

$$f''_{xy} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$$

$$f''_{yy} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right)$$

$$f''_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right).$$

Example 16

- 1 Compute all the second order partial derivatives of the function $z = \arctan(x^2y)$.
- 2 Given $z = y \cos \frac{y}{x}$. Compute z''_{xy} and z''_{yx} .
- 3 Given the function $f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$ Compute $f''_{xy}(0, 0)$ and $f''_{yx}(0, 0)$.

Theorem 1.7

Assume that $z = f(x, y)$ has partial derivatives f''_{xy} and f''_{yx} near $M_0(x_0, y_0)$ which are *continuous at M_0* . Then,

$$f''_{xy}(M_0) = f''_{yx}(M_0).$$

Definition 6

Second order total differential

$$d^2z = d(dz) = \frac{\partial^2 f}{\partial x^2} (dx)^2 + 2 \frac{\partial^2 f}{\partial x \partial y} dx dy + \frac{\partial^2 f}{\partial y^2} (dy)^2.$$

Higher order total differential of $z = f(x, y)$:

$$d^n z = d(d^{n-1} z), \quad n \geq 3.$$

Example 17

- ❶ Given $z = \arctan(xy)$. Compute the second order differential d^2z and $d^2z(1, -1)$.
- ❷ Given $z = 2^x \sin y$. Compute the second order differential d^2z and $d^2z(0, \pi)$.

$$\text{❶ } z'_x = \frac{y}{1+x^2y^2}; z'_y = \frac{x}{1+x^2y^2}; z''_{xx} = -\frac{2xy^3}{(1+x^2y^2)^2}$$

$$z''_{xy} = \frac{1-x^2y^2}{(1+x^2y^2)^2} = z''_{yx}; z''_{yy} = -\frac{2yx^3}{(1+x^2y^2)^2}.$$

Hence,

$$d^2z = \frac{-2xy^3d^2x + 2(1-x^2y^2)dxdy - 2yx^3d^2y}{(1+x^2y^2)^2}$$

$$d^2z(1, -1) = \frac{d^2x + d^2y}{2}$$

Theorem 1.8

Let $f(x, y)$ have continuous partial derivatives up to order $n + 1$ in an open set $U \subset \mathbb{R}^2$, $M_0(x_0, y_0), M(x_0 + \Delta x, y_0 + \Delta y) \in U$. There exists $0 < \theta < 1$ such that

$$f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) = \left(df(x_0, y_0) + \frac{1}{2!} d^2 f(x_0, y_0) + \dots + \frac{1}{n!} d^n f(x_0, y_0) \right) + \frac{1}{(n+1)!} d^{n+1} f(x_0 + \theta \Delta x, y_0 + \theta \Delta y).$$

$n = 2$, we have the explicit expansion as follows:

$$\begin{aligned} f(x, y) = & f(x_0, y_0) + f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0) + \\ & + \frac{1}{2} [f''_{xx}(x_0, y_0)(x - x_0)^2 + 2f''_{xy}(x_0, y_0)(x - x_0)(y - y_0) + f''_{yy}(x_0, y_0)(y - y_0)^2] + \\ & + R_2(x - x_0, y - y_0). \end{aligned}$$

The sum

$$\begin{aligned} & f(x_0, y_0) + f'_x(x_0, y_0)(x - x_0) + f'_y(x_0, y_0)(y - y_0) + \\ & + \frac{1}{2} [f''_{xx}(x_0, y_0)(x - x_0)^2 + 2f''_{xy}(x_0, y_0)(x - x_0)(y - y_0) + f''_{yy}(x_0, y_0)(y - y_0)^2] \end{aligned}$$

is called *the second degree Taylor polynomial* of the function $f(x, y)$

Example 18

Compute the second degree Taylor polynomial of the function $f(x, y) = \arctan(xy)$ at $(1; -1)$.

From the previous slide, we obtain the second degree Taylor polynomial:

$$T_2(x, y) = -\frac{\pi}{4} - \frac{x-1}{2} + \frac{y+1}{2} + \frac{1}{4}((x-1)^2 + (y+1)^2).$$

Definition 7

*The directional derivative of $z = f(x, y)$ at (x_0, y_0) in the direction given by a **unit vector** $\mathbf{u} = (a, b)$ is*

$$\begin{aligned} D_{\mathbf{u}}f(x_0, y_0) &= \lim_{h \rightarrow 0} \frac{f((x_0, y_0) + h\mathbf{u}) - f(x_0, y_0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x_0 + ha, y_0 + hb) - f(x_0, y_0)}{h}. \end{aligned}$$

$D_{\mathbf{u}}f(x_0, y_0)$ shows the rate of change of z at (x_0, y_0) in the direction of $\mathbf{u}$.

Special cases:

- $\mathbf{u} = (1, 0)$: $D_{\mathbf{u}}f = \frac{\partial u}{\partial x}$.
- $\mathbf{u} = (0, 1)$: $D_{\mathbf{u}}f = \frac{\partial u}{\partial y}$.

Theorem 1.9

If $f(x, y)$ is differentiable function, $\mathbf{u} = (a, b)$, then the directional derivative can be calculated as follows

$$D_{\mathbf{u}}f(x, y) = f'_x \cdot a + f'_y \cdot b.$$

Definition 8

The gradient vector of $f(x, y)$ is defined and denoted as follows

$$\nabla f(x, y) = \overrightarrow{\text{grad}} f = (f'_x(x, y), f'_y(x, y)) = f'_x \overrightarrow{\mathbf{i}} + f'_y \overrightarrow{\mathbf{j}},$$

where $\overrightarrow{\mathbf{i}}, \overrightarrow{\mathbf{j}}$ are the unit vectors of the coordinate axes.

$$D_{\mathbf{u}} f(x, y) = \nabla f \cdot \mathbf{u}.$$

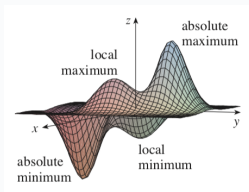
Theorem 1.10

For any unit vector $\mathbf{u}$, the directional derivative $D_{\mathbf{u}} f$ attains the maximum value $|\nabla f(x, y)|$ when $\mathbf{u}$ has the same direction as the gradient vector ∇f .

Definition 9

Given $z(x, y): D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$. $B(M_0, \varepsilon) \subset D$. We say that

- $z(x, y)$ has a **local/relative maximum** at M_0 if $z(M) \leq z(M_0) \forall M \in B(M_0; \varepsilon)$. $f(M_0)$: **local maximum value**.
- $z(x, y)$ has a **local/relative minimum** at M_0 if $z(M) \geq z(M_0) \forall M \in B(M_0; \varepsilon)$. $f(M_0)$: **local minimum value**.
- $z(x, y)$ has an **absolute/global maximum** at M_0 if $z(M) - z(M_0) \leq 0 \forall M \in D$.
- $z(x, y)$ has an **absolute/global minimum** at M_0 if $z(M) - z(M_0) \geq 0 \forall M \in D$.



Theorem 1.11 (First derivative test)

If $z(x, y)$ has a local maximum or local minimum at M_0 and there exist $z'_x(M_0), z'_y(M_0)$, then $z'_x(M_0) = z'_y(M_0) = 0$.

- M_0 is a local extreme point of $z(x, y)$, in particular x_0 is a local extreme point of $z(x, y_0)$.
- By Fermat's theorem $z'_x(M_0) = 0$. Similarly, $z'_y(M_0) = 0$.

M_0 : stationary point.

Theorem 1.12 (Second derivative test)

Suppose the second order partial derivatives of $z(x, y)$ are continuous in $B(M_0; \varepsilon)$, and $z'_x(M_0) = z'_y(M_0) = 0$. Denote

$$A = z''_{xx}(M_0), B = z''_{xy}(M_0), C = z''_{yy}(M_0), \Delta = B^2 - AC.$$

- If $\Delta < 0$:
 - ▶ $A > 0$ then z attains a local minimum at M_0 ,
 - ▶ $A < 0$ then z attains a local maximum at M_0 .
- If $\Delta > 0$, $z(x, y)$ does not attain a local extreme value at M_0 (M_0 is a saddle point).
- If $\Delta = 0$: no conclusion can be drawn.

Example 19

Find the local maximum and minimum values of the functions

❶ $z = x^2 - 3xy + x + y^3 - 2.$

❷ $z = e^{-y}(3x - x^3 - y).$

❸ $z = x^4 + y^4 + (x + y)^3.$

Theorem 1.13 (Extreme value theorem)

A continuous function $f: D \subset \mathbb{R}^n \rightarrow \mathbb{R}$ on a closed, bounded domain D attains its absolute maximum and minimum values on that domain.

- 1 Evaluate the values of f at critical points in D .
- 2 Find the extreme values of f on the boundary of D .
- 3 The largest of these values is the absolute maximum value of f , the smallest of these values is the absolute minimum value of f .

Example 20

Find the absolute maximum and minimum values of the following functions

- 1 $z = x^2 + 2xy - 4x + 8y$ on the rectangle enclosed by $x = 0, x = 1, y = 0, y = 2$.
- 2 $z = x^2 - y^2$ on the domain $x^2 + y^2 \leq 4$.
- 3 $z = xy(3 - x - y)$ on the domain $0 \leq x \leq 2, 0 \leq y \leq 2$.

An equation $F(x, y) = 0$ defines some functions $y = f(x)$ which are called **implicit functions**.
By definition, $F(x, y(x)) = 0$. Differentiating both sides with respect to x , we obtain

$$F'_x + F'_y \cdot y'(x) = 0 \xrightarrow{F'_y \neq 0} y'(x) = -\frac{F'_x}{F'_y}.$$